

Sl.no	Title	Author & Year	Journal / Conference	Methodology	Merits	Demerits (Limitations)	Dataset Source	Remarks	Reference Link
1	A Formal Security Analysis of the Signal Messaging Protocol	Cohn-Gordon et al., 2017	IEEE EuroS&P	Formal cryptographic proof, protocol analysis	Proven forward secrecy and post-compromise security	Depends on phone-number identity and centralized servers	Simulated protocol traces	Foundation for modern secure messaging	https://ieeexplore.ieee.org/document/7961973
2	The X3DH Key Agreement Protocol	Marlinspike & Perrin, 2016	Signal Whitepaper	Protocol design and security reasoning	Supports asynchronous secure messaging	Metadata leakage via prekey servers	No public dataset	Suitable for offline-first systems	https://signal.org/docs/specifications/x3dh/
3	Curve25519: New Diffie–Hellman Speed Records	Bernstein, 2006	PKC (Springer)	Mathematical design and benchmarking	Fast, side-channel resistant, small keys	Does not define identity or trust model	Mathematical benchmarks	Ideal for military-grade crypto	https://cr.yp.to/ecdh/curve25519-20060209.pdf
4	The Galois/Counter Mode of Operation (GCM)	McGrew & Viega, 2004	NIST Workshop	Cryptographic construction and evaluation	Efficient authenticated encryption	Nonce reuse breaks security	Test vectors	Used in AES-GCM military systems	https://nvlpubs.nist.gov
5	Backdoored Hash Functions: Immunizing HMAC and HKDF	Fischlin et al., 2014	EUROCRYPT	Theoretical cryptographic analysis	Resilient to cryptographic backdoors	Applies mainly to keyed hashes	Theoretical models	Justifies HKDF usage	https://link.springer.com/chapter/10.1007/978-3-642-55220-5_7
6	New Directions in Cryptography	Diffie & Hellman, 1976	IEEE	Mathematical protocol design	Introduced public-key cryptography	Vulnerable to MITM without auth	Conceptual	Foundation of secure messaging	https://ieeexplore.ieee.org/document/1055638
7	Probabilistic Encryption	Goldwasser & Micali, 1982	STOC	Formal cryptographic proof	Introduced semantic security	Abstract, not system-focused	Theoretical	Basis of modern encryption	https://dl.acm.org/doi/10.1145/800070.802212
8	Untraceable Electronic Mail	David Chaum, 1981	CACM	Protocol design (Mix networks)	Strong anonymity	High latency, poor real-time use	Simulated traffic	Influences anonymity systems	https://dl.acm.org/doi/10.1145/358549.358563
9	The Economics of Anonymous Communications	Danezis & Anderson, 2003	PET	Adversarial and economic modeling	Highlights traffic analysis risks	No concrete protocol	Analytical models	Important for metadata threats	https://www.cl.cam.ac.uk/~rja14/Papers/econ.pdf
10	On End-to-End Encryption	Hale & Komlo, 2019	IEEE S&P Workshops	Formal definition and case studies	Clarifies true E2EE meaning	No implementation	Case studies	Supports zero-trust design	https://ieeexplore.ieee.org/document/8835340

11	SoK: Authentication Ceremonies in Secure Messaging	Alatawi & Saxena, 2023	WiSec	Comparative system analysis	Identifies MitM risks	Limited design solutions	App comparisons	Relevant for military auth	https://dl.acm.org/doi/10.1145/3558482.3581773
12	Peeking Into the Black Box: Understanding E2EE	Schaewitz et al., 2021	EuroUSEC	Qualitative interviews	Reveals user misunderstanding	Small sample size	Interview data	Shows usability gaps	https://www.ndss-symposium.org
13	Secure Messaging for Military Applications	Alshaer et al., 2022	Military Comm. Conf.	Architecture design + threat modeling	Tailored for hostile environments	Limited civilian applicability	Simulated military data	Direct military relevance	https://ieeexplore.ieee.org
14	Offline Secure Messaging in Disconnected Environments	Patil et al., 2020	Elsevier FGCS	Store-and-forward architecture	Works without network	Increased latency	Network simulations	Supports offline storage	https://www.sciencedirect.com
15	Post-Compromise Secure Messaging	Cohn-Gordon et al., 2016	IEEE CSF	Formal protocol verification	Recovery after compromise	Higher protocol complexity	Formal models	Critical for device seizure	https://ieeexplore.ieee.org